

A Technical Introduction to Chain of thought reasoning

Section 1

Artist styles Some text-to-image models are capable of imitating the style of particular artists by name. For example, the phrase in the style of Greg Rutkowski has been used in Stable Diffusion and Midjourney prompts to generate images in the distinctive style of Polish digital artist Greg Rutkowski. Famous artists such as Vincent van Gogh and Salvador Dalí have also been used for styling and testing.

Section 2

Using language models to generate prompts LLMs themselves can be used to compose prompts for LLMs. The automatic prompt engineer algorithm uses one LLM to beam search over prompts for another LLM:

Section 3

There are two LLMs. One is the target LLM, and another is the prompting LLM. Prompting LLM is presented with example input-output pairs, and asked to generate instructions that could have caused a model following the instructions to generate the outputs, given the inputs. Each of the generated instructions is used to prompt the target LLM, followed by each of the inputs. The log-probabilities of the outputs are computed and added. This is the score of the instruction. The highest-scored instructions are given to the prompting LLM for further variations. Repeat until some stopping criteria is reached, then output the highest-scored instructions. CoT examples can be generated by LLM themselves. In "auto-CoT", a library of questions are converted to vectors by a model such as BERT. The question vectors are clustered. Questions close to the centroid of each cluster are selected, in order to have a subset of diverse questions. An LLM does zero-shot CoT on each selected question. The question and the corresponding CoT answer are added to a dataset of demonstrations. These diverse demonstrations can then be added to prompts for few-shot learning.

Section 4

GraphRAG (coined by Microsoft Research) is a technique that extends RAG with the use of a knowledge graph to allow the model to connect disparate pieces of information, synthesize insights, and understand summarized semantic concepts over large data collections. It was shown to be effective on datasets like the Violent Incident Information from News Articles.

Section 5

Prompt injection is a cybersecurity exploit in which adversaries craft inputs that appear legitimate but are

designed to cause unintended behavior in machine learning models, particularly large language models. This attack takes advantage of the model's inability to distinguish between developer-defined prompts and user inputs, allowing adversaries to bypass safeguards and influence model behaviour. While LLMs are designed to follow trusted instructions, they can be manipulated into carrying out unintended responses through carefully crafted inputs.

Section 6

Terminology The Oxford English Dictionary defines prompt engineering as "The action or process of formulating and refining prompts for an artificial intelligence program, algorithm, etc., in order to optimize its output or to achieve a desired outcome; the discipline or profession concerned with this." In 2023, prompt ("an instruction given to an artificial intelligence program, algorithm, etc., which determines or influences the content it generates") was the runner-up to Oxford's word of the year.

Section 7

Retrieval-augmented generation (RAG) Retrieval-augmented generation is a technique that enables GenAI models to retrieve and incorporate new information. It modifies interactions with an LLM so that the model responds to user queries with reference to a specified set of documents, using this information to supplement information from its pre-existing training data. This allows LLMs to use domain-specific and/or updated information. RAG improves large language models by incorporating information retrieval before generating responses. Unlike traditional LLMs that rely on static training data, RAG pulls relevant text from databases, uploaded documents, or web sources. By dynamically retrieving information, RAG enables AI to generate more accurate responses and fewer AI hallucinations without frequent retraining.

Section 8

Tree-of-thought Tree-of-thought prompting generalizes chain-of-thought by generating multiple lines of reasoning in parallel, with the ability to backtrack or explore other paths. It can use tree search algorithms like breadth-first, depth-first, or beam.

Section 9

Textual inversion and embeddings For text-to-image models, textual inversion performs an optimization process to create a new word embedding based on a set of example images. This embedding vector acts as a "pseudo-word" which can be included in a prompt to express the content or style of the examples.

Section 10

Techniques Common terms used to describe various specific prompt engineering techniques include chain-of-thought, tree-of-thought, and retrieval-augmented generation (RAG). A 2024 survey of the field identified over 50 distinct text-based prompting techniques, 40 multimodal variants, and a vocabulary of 33 terms used across prompting research, highlighting a present lack of standardised terminology for prompt engineering. Vibe coding is an AI-assisted software development method where a user prompts an LLM with a description of what they want and lets it generate or edit the code. In 2025, "vibe coding" was the Collins Dictionary word of the year.

Section 11

Prompting to estimate model sensitivity Research consistently demonstrates that LLMs are highly sensitive to subtle variations in prompt formatting, structure, and linguistic properties. Some studies have shown up to 76 accuracy points across formatting changes in few-shot settings. Linguistic features significantly influence prompt effectiveness—such as morphology, syntax, and lexico-semantic changes—which meaningfully enhance task performance across a variety of tasks. Clausal syntax, for example, improves consistency and reduces uncertainty in knowledge retrieval. This sensitivity persists even with larger model sizes, additional few-shot examples, or instruction tuning. To address sensitivity of models and make them more robust, several evaluative methods have been proposed. FormatSpread facilitates systematic analysis by evaluating a range of plausible prompt formats, offering a more comprehensive performance interval. Similarly, PromptEval estimates performance distributions across diverse prompts, enabling robust metrics such as performance quantiles and accurate evaluations under constrained budgets.

Section 12

Rationale Research has found that the performance of large language models (LLMs) is highly sensitive to choices such as the ordering of examples, the quality of demonstration labels, and even small variations in phrasing. In some cases, reordering examples in a prompt produced accuracy shifts of more than 40 percent.

Section 13

Image prompting In 2023, Meta's AI research released Segment Anything, a computer vision model that can perform image segmentation by prompting. As an alternative to text prompts, Segment Anything can accept bounding boxes, segmentation masks, and foreground/background points.